

Lab 32 — Pandas Data Frame Creation and Combining

Aim

To create Pandas DataFrames from dictionaries and combine them using concatenation and merging.

Algorithm / Procedure

1. Import the Pandas library.
2. Create DataFrames using Python dictionaries.
3. Use `pd.concat()` to combine DataFrames by rows.
4. Create another DataFrame containing grades.
5. Use `pd.merge()` to join DataFrames using the common name column.
6. Display the concatenated and merged DataFrames.

Program

```
import pandas as pd

df1=pd.DataFrame({'name':['A','B','C'],'marks':[80,90,70]})

df2=pd.DataFrame({'name':['D'],'marks':[85]})

combined=pd.concat([df1,df2],ignore_index=True)

print("Concatenated:\n",combined)

df3=pd.DataFrame({'name':['A','B','C'],'grade':['A','A+','B']})

merged=pd.merge(df1,df3,on='name',how='inner')

print("Merged:\n",merged)
```

Output

Concatenated:

	name	marks
0	A	80
1	B	90
2	C	70
3	D	85

Merged:

	name	marks	grade
0	A	80	A
1	B	90	A+
2	C	70	B

Result

Pandas DataFrames were successfully created, concatenated, and merged using `pd.concat()` and `pd.merge()` on a common key.